

Assignment – 2 – Aastha Thakker (004) – MSc CS

Nmap Scanning

A) Target Specification

1. Scanning a domain name.

```
aastha@Aastha:~$ nmap aasthathakker.com
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-31 16:40 IST
Nmap scan report for aasthathakker.com (185.230.63.171)
Host is up (0.29s latency).
Other addresses for aasthathakker.com (not scanned): 185.230.63.186 185.230.63.107
rDNS record for 185.230.63.171: unallocated.63.wixsite.com
Not shown: 990 filtered tcp ports (no-response), 6 filtered tcp ports (host-unreach)
PORT      STATE SERVICE
80/tcp    open  http
82/tcp    open  xfer
443/tcp   open  https
445/tcp   open  microsoft-ds
```

2. Scanning a range of IP

```
aastha@Aastha:~$ nmap 192.168.178.1-254
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-31 16:54 IST
Nmap scan report for 192.168.178.131
Host is up (0.00012s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    open  smtp
80/tcp    open  http

Nmap done: 254 IP addresses (1 host up) scanned in 29.85 seconds
```

3. Creating a file and scanning all domains/lps present in that file.

```
(aastha@kali)-[~]  
$ cat scan.txt  
google.com  
aasthathakker.com  
whereismymetro.com
```

```
(aastha@kali)-[~]  
$ nmap -iL scan.txt  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-27 04:27 EST  
Nmap scan report for google.com (142.250.67.78)  
Host is up (0.0013s latency).  
Other addresses for google.com (not scanned): 2404:6800:4002:806::200e  
rDNS record for 142.250.67.78: maa05s13-in-f14.1e100.net  
Not shown: 994 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
23/tcp    open  telnet  
80/tcp    open  http  
113/tcp   closed ident  
443/tcp   open  https  
1000/tcp  open  cadlock  
  
Nmap scan report for aasthathakker.com (185.230.63.186)  
Host is up (0.00068s latency).  
Other addresses for aasthathakker.com (not scanned): 185.230.63.171 185.230.63.107  
rDNS record for 185.230.63.186: unallocated.63.wixsite.com  
Not shown: 994 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
23/tcp    open  telnet  
80/tcp    open  http  
113/tcp   closed ident  
443/tcp   open  https  
1000/tcp  open  cadlock  
  
Nmap scan report for whereismymetro.com (216.198.79.1)  
Host is up (0.00074s latency).  
Not shown: 994 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
23/tcp    open  telnet  
80/tcp    open  http  
113/tcp   closed ident  
443/tcp   open  https  
1000/tcp  open  cadlock  
  
Nmap done: 3 IP addresses (3 hosts up) scanned in 15.97 seconds
```

B) Nmap Scan Techniques

1. TCP SYN port scan (Default)

```

C:\Users\Admin>nmap 185.230.63.186 -sS
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 12:14 +0530
Nmap scan report for 186.63.230.185.in-addr.arpa (185.230.63.186)
Host is up (0.27s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
43/tcp    closed whois
80/tcp    open  http
82/tcp    open  xfer
83/tcp    closed mit-ml-dev
443/tcp   open  https
445/tcp   open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 19.00 seconds

```

2. TCP connect port scan (Default without root privilege)

```

C:\Users\Admin>nmap 185.230.63.186 -sT
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 12:16 +0530
Nmap scan report for 186.63.230.185.in-addr.arpa (185.230.63.186)
Host is up (0.26s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
43/tcp    closed whois
80/tcp    open  http
82/tcp    open  xfer
83/tcp    closed mit-ml-dev
443/tcp   open  https
445/tcp   open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 37.21 seconds

```

3. UDP port scan

```

C:\Users\Admin>nmap 185.230.63.186 -sU
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 12:45 +0530
Nmap scan report for unallocated.63.wixsite.com (185.230.63.186)
Host is up (0.25s latency).
All 1000 scanned ports on unallocated.63.wixsite.com (185.230.63.186) are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 250.70 seconds

```

4. TCP ACK port scan

```

C:\Users\Admin>nmap 185.230.63.186 -sA
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 12:52 +0530
Nmap scan report for 186.63.230.185.in-addr.arpa (185.230.63.186)
Host is up (0.28s latency).
All 1000 scanned ports on 186.63.230.185.in-addr.arpa (185.230.63.186) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 22.59 seconds

```

C) Host Discovery

1. Disable port scanning. Host discovery only.

```
PS C:\Users\Admin> nmap -sn 185.230.63.171
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:08 +0530
Nmap scan report for unallocated.63.wixsite.com (185.230.63.171)
Host is up (0.25s latency).
Nmap done: 1 IP address (1 host up) scanned in 1.45 seconds
```

2. Disable host discovery. Port scan only.

```
PS C:\Users\Admin> nmap -Pn 185.230.63.171
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:09 +0530
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.27s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
43/tcp    closed whois
80/tcp    open  http
82/tcp    open  xfer
83/tcp    closed mit-ml-dev
443/tcp   open  https
445/tcp   open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 30.16 seconds
```

3. Never do DNS resolution.

```
PS C:\Users\Admin> nmap -n 185.230.63.171
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:12 +0530
Nmap scan report for 185.230.63.171
Host is up (0.28s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
43/tcp    closed whois
80/tcp    open  http
82/tcp    open  xfer
83/tcp    closed mit-ml-dev
443/tcp   open  https
445/tcp   open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 20.51 seconds
```

D) Port Specification

1. Port scan for port x

```
PS C:\Users\Admin> nmap 185.230.63.171 -p 21
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:13 +0530
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.35s latency).

PORT      STATE      SERVICE
21/tcp    filtered  ftp

Nmap done: 1 IP address (1 host up) scanned in 5.09 seconds
PS C:\Users\Admin> nmap 185.230.63.171 -p 3388
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:13 +0530
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.33s latency).

PORT      STATE      SERVICE
3388/tcp  filtered  cbserver

Nmap done: 1 IP address (1 host up) scanned in 4.18 seconds
PS C:\Users\Admin> nmap 185.230.63.171 -p 8080
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:14 +0530
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.35s latency).

PORT      STATE      SERVICE
8080/tcp  filtered  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 4.31 seconds
```

2. Port range

```
PS C:\Users\Admin> nmap 185.230.63.171 -p 20-110
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:14 +0530
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.28s latency).
Not shown: 87 filtered tcp ports (no-response)
PORT      STATE      SERVICE
43/tcp    closed    whois
80/tcp    open      http
82/tcp    open      xfer
83/tcp    closed    mit-ml-dev

Nmap done: 1 IP address (1 host up) scanned in 7.28 seconds
```

3. Port scans all ports.

```

PS C:\Users\Admin> nmap 185.230.63.171 -p-
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:15 +0530
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.26s latency).
Not shown: 65525 filtered tcp ports (no-response), 1 filtered tcp ports (admin-prohibited)
PORT      STATE SERVICE
43/tcp    closed whois
80/tcp    open  http
82/tcp    open  xfer
83/tcp    closed mit-ml-dev
443/tcp   open  https
445/tcp   open  microsoft-ds
446/tcp   closed ddm-rdb
447/tcp   closed ddm-dfm
10443/tcp closed cirrossp

Nmap done: 1 IP address (1 host up) scanned in 534.85 seconds

```

4. Port scan from service name

```

C:\Users\Admin>nmap 185.230.63.171 -p http,https
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:17 +0530
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.25s latency).

PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
8008/tcp  filtered http

Nmap done: 1 IP address (1 host up) scanned in 4.13 seconds

```

5. Fast port scan (100 ports)

```

C:\Users\Admin>nmap 185.230.63.171 -F
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:17 +0530
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.25s latency).
Not shown: 97 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
445/tcp   open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 6.89 seconds

```

6. Port scans the top x ports

```

C:\Users\Admin>nmap 185.230.63.171 --top-ports 2000
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:18 +0530
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.25s latency).
Not shown: 1993 filtered tcp ports (no-response)
PORT      STATE SERVICE
43/tcp    closed whois
80/tcp    open  http
82/tcp    open  xfer
83/tcp    closed mit-ml-dev
443/tcp   open  https
445/tcp   open  microsoft-ds
447/tcp   closed ddm-dfm

Nmap done: 1 IP address (1 host up) scanned in 25.21 seconds

```

E) Service & Version Detection

1. Attempts to determine the version of the service running on port.

```

C:\Users\Admin>nmap 185.230.63.171 -sV
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:20 +0530
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.27s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
43/tcp    closed whois
80/tcp    open  http
82/tcp    open  xfer?
83/tcp    closed mit-ml-dev
443/tcp   open  ssl/https?
445/tcp   open  microsoft-ds?
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint
at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port80-TCP:V=7.98%I=7%D=2/1%Time=697F05CF%P=i686-pc-windows-windows%r(G
SF:etRequest,79,"HTTP/1.0\x20403\x20Forbidden\r\nX-Seen-By:\x20xIKq3Iotbb
SF:Lp4\+7DTTMx8R9slopJdhD\+WySraMrpIY8=\r\nConnection:\x20close\r\nContent
SF:-Length:\x200\r\n\r\n")%r(HTTPOptions,79,"HTTP/1.0\x20403\x20Forbidden
SF:\r\nX-Seen-By:\x20xIKq3IotbbLp4\+7DTTMx8R9slopJdhD\+WySraMrpIY8=\r\nCon
SF:nection:\x20close\r\nContent-Length:\x200\r\n\r\n")%r(RTSPRequest,79,"H
SF:TTP/1.0\x20403\x20Forbidden\r\nX-Seen-By:\x20xIKq3IotbbLp4\+7DTTMx8R9s
SF:lopJdhD\+WySraMrpIY8=\r\nConnection:\x20close\r\nContent-Length:\x200\r
SF:\n\r\n")%r(FourOhFourRequest,79,"HTTP/1.0\x20403\x20Forbidden\r\nX-See
SF:n-By:\x20xIKq3IotbbLp4\+7DTTMx8R9slopJdhD\+WySraMrpIY8=\r\nConnection:\
SF:x20close\r\nContent-Length:\x200\r\n\r\n");

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 70.14 seconds

```

2. Enable light mode. Lower possibility of correctness. Faster

```

C:\Users\Admin>nmap 185.230.63.171 -sV --version-light
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:22 +0530
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.25s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
43/tcp    closed whois
80/tcp    open  http?
82/tcp    open  xfer?
83/tcp    closed mit-ml-dev
443/tcp   open  ssl/https?
445/tcp   open  microsoft-ds?
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port80-TCP:V=7.98%I=2%D=2/1%Time=697F0658%P=i686-pc-windows-windows%r(G
SF:etRequest,79,"HTTP/1\0\0\20403\20Forbidden\r\nX-Seen-By:\x20T7xPrjRFKD
SF:MHVv938PVVfx9slopJdhD\+WySraMrpIY8=\r\nConnection:\x20close\r\nContent-
SF:Length:\x200\r\n\r\n")%r(HTTPOptions,79,"HTTP/1\0\0\20403\20Forbidden\
SF:r\nX-Seen-By:\x20dwc60INy8NFddnU/0WdLOB9slopJdhD\+WySraMrpIY8=\r\nConne
SF:ction:\x20close\r\nContent-Length:\x200\r\n\r\n")%r(RTSPRequest,79,"HTT
SF:P/1\0\0\20403\20Forbidden\r\nX-Seen-By:\x209WD8GAcPjgs/NglWkD2i0h9slop
SF:JdhD\+WySraMrpIY8=\r\nConnection:\x20close\r\nContent-Length:\x200\r\n\
SF:r\n")%r(FourOhFourRequest,79,"HTTP/1\0\0\20403\20Forbidden\r\nX-Seen-B
SF:y:\x20dwc60INy8NFddnU/0WdLOB9slopJdhD\+WySraMrpIY8=\r\nConnection:\x20c
SF:lose\r\nContent-Length:\x200\r\n\r\n");

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 44.73 seconds

```

3. Enable intensity level 9. Higher possibility of correctness. Slower

```

C:\Users\Admin>nmap 185.230.63.171 -sV --version-all
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:24 +0530
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.26s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
43/tcp    closed whois
80/tcp    open  http
82/tcp    open  xfer?
83/tcp    closed mit-ml-dev
443/tcp   open  ssl/https?
445/tcp   open  microsoft-ds?
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.c
gi?new-service :
SF-Port80-TCP:V=7.98%I=9%D=2/1%Time=697F06D1%P=i686-pc-windows-windows%r(G
SF:etRequest,79,"HTTP/1\0\0\20403\20Forbidden\r\nX-Seen-By:\x20ui2Pjuc3zC
SF:IHEw72qKeyh9slopJdhD\+WySraMrpIY8=\r\nConnection:\x20close\r\nContent-
SF:Length:\x200\r\n\r\n")%r(HTTPOptions,79,"HTTP/1\0\0\20403\20Forbidden\
SF:r\nX-Seen-By:\x20dwc60INy8NFddnU/0WdLOB9slopJdhD\+WySraMrpIY8=\r\nConne

```

4. Enables OS detection, version detection, script scanning, and traceroute


```

(aastha@kali)-[~]
$ nmap -sF aasthathakker.com
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-27 05:08 EST
Stats: 0:01:51 elapsed; 0 hosts completed (1 up), 1 undergoing FIN Scan
FIN Scan Timing: About 53.50% done; ETC: 05:12 (0:01:37 remaining)
Nmap scan report for aasthathakker.com (185.230.63.171)
Host is up (0.20s latency).
Other addresses for aasthathakker.com (not scanned): 185.230.63.107 185.230.63.186
rDNS record for 185.230.63.171: unallocated.63.wixsite.com
All 1000 scanned ports on aasthathakker.com (185.230.63.171) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 206.74 seconds

```

-sN (NULL Scan): This sends a TCP packet with **no flags set**.

Same logic: Closed → RST response, Open → no response

Used for: Stealth probing and Firewall rule testing

```

(aastha@kali)-[~]
$ nmap -sN aasthathakker.com
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-27 05:09 EST
Nmap scan report for aasthathakker.com (185.230.63.171)
Host is up (0.19s latency).
Other addresses for aasthathakker.com (not scanned): 185.230.63.107 185.230.63.186
rDNS record for 185.230.63.171: unallocated.63.wixsite.com
All 1000 scanned ports on aasthathakker.com (185.230.63.171) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 197.96 seconds

```

-sX (Xmas Scan): This sets **FIN, PSH, and URG flags** together. It's called "Xmas" because the packet is "lit up" like a Christmas tree.

Logic: Closed → RST, Open → no response

Used for: Detecting firewall behavior and OS fingerprinting differences

```

(aastha@kali)-[~]
$ nmap -sX aasthathakker.com
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-27 05:13 EST
Nmap scan report for aasthathakker.com (185.230.63.186)
Host is up (0.20s latency).
Other addresses for aasthathakker.com (not scanned): 185.230.63.171 185.230.63.107
rDNS record for 185.230.63.186: unallocated.63.wixsite.com
All 1000 scanned ports on aasthathakker.com (185.230.63.186) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 16.29 seconds

```

Switch	Description	Technical Detail
-sS	TCP SYN Scan (Default)	Performs a "half-open" scan; it is stealthy as it doesn't complete the TCP connection. +1
-sT	TCP Connect Scan	Completes the 3-way handshake; used when the user lacks root privileges. +1
-sU	UDP Port Scan	Scans for open UDP ports (e.g., DNS, SNMP). +1
-sA	TCP ACK Scan	Used to map out firewall rulesets; identifies filtered vs. unfiltered ports. +1
-sM	Maimon Scan	Sets FIN and ACK flags to observe the target's response. +1

F) OS Detection

1. Remote OS detection using TCP/IP stack fingerprinting

```
PS C:\Users\Admin> nmap 185.230.63.171 -O
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:31 +0530
Nmap scan report for unallocated.63.wixsite.com (185.230.63.171)
Host is up (0.27s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
43/tcp    closed whois
80/tcp    open  http
82/tcp    open  xfer
83/tcp    closed mit-ml-dev
443/tcp   open  https
445/tcp   open  microsoft-ds
OS fingerprint not ideal because: Didn't receive UDP response. Please try again with -sSU
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 27.02 seconds
```

2. Makes Nmap guess more aggressively

```
PS C:\Users\Admin> nmap 185.230.63.171 --osscan-guess
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:33 +0530
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.27s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
43/tcp    closed whois
80/tcp    open  http
82/tcp    open  xfer
83/tcp    closed mit-ml-dev
443/tcp   open  https
445/tcp   open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 21.99 seconds
```

G) Output

1. Normal output to the file normal.file

```
(aastha@kali)-[~]
$ sudo nmap 185.230.63.186 -oN filename
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-27 05:46 EST
Nmap scan report for unallocated.63.wixsite.com (185.230.63.186)
Host is up (0.21s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE      SERVICE
80/tcp    open      http
82/tcp    open      xfer
443/tcp   open      https
445/tcp   filtered  microsoft-ds
2000/tcp  open      cisco-sccp
5060/tcp  open      sip

Nmap done: 1 IP address (1 host up) scanned in 9.88 seconds

(aastha@kali)-[~]
$ ls
Desktop  Documents  Downloads  filename  Music  Pictures  Public  scan.txt  Templates  Videos
```

```
(aastha@kali)-[~]
$ sudo nmap 185.230.63.186 --reason -oN filename
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-27 05:47 EST
Nmap scan report for unallocated.63.wixsite.com (185.230.63.186)
Host is up, received syn-ack ttl 238 (0.24s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE      SERVICE      REASON
80/tcp    open      http         syn-ack ttl 238
82/tcp    open      xfer         syn-ack ttl 238
443/tcp   open      https        syn-ack ttl 240
445/tcp   filtered  microsoft-ds no-response
2000/tcp  open      cisco-sccp   syn-ack ttl 63
5060/tcp  open      sip          syn-ack ttl 63

Nmap done: 1 IP address (1 host up) scanned in 12.89 seconds
```

2. Output in the three major formats at once

```
C:\Users\Admin>nmap 185.230.63.171 -oA results
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:40 +0530
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.29s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE      SERVICE
43/tcp    closed    whois
80/tcp    open      http
82/tcp    open      xfer
83/tcp    closed    mit-ml-dev
443/tcp   open      https
445/tcp   open      microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 17.15 seconds
```

```

02/01/2026 01:41 PM 488 results.gnmap
02/01/2026 01:41 PM 496 results.nmap
02/01/2026 01:41 PM 9,886 results.xml

```

3. Increase the verbosity level (use -vv or more for greater effect)

```

C:\Users\Admin>nmap 185.230.63.171 -v
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:39 +0530
Initiating Ping Scan at 13:39
Scanning 185.230.63.171 [4 ports]
Completed Ping Scan at 13:39, 0.32s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 13:39
Completed Parallel DNS resolution of 1 host. at 13:39, 0.01s elapsed
Initiating SYN Stealth Scan at 13:39
Scanning 171.63.230.185.in-addr.arpa (185.230.63.171) [1000 ports]
Discovered open port 445/tcp on 185.230.63.171
Discovered open port 80/tcp on 185.230.63.171
Discovered open port 443/tcp on 185.230.63.171
Discovered open port 82/tcp on 185.230.63.171
Completed SYN Stealth Scan at 13:39, 19.01s elapsed (1000 total ports)
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up (0.27s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
43/tcp    closed whois
80/tcp    open  http
82/tcp    open  xfer
83/tcp    closed mit-ml-dev
443/tcp   open  https
445/tcp   open  microsoft-ds

Read data files from: C:\Program Files (x86)\Nmap
Nmap done: 1 IP address (1 host up) scanned in 20.53 seconds
Raw packets sent: 2007 (88.284KB) | Rcvd: 31 (1.332KB)

C:\Users\Admin>nmap 185.230.63.171 -vvv
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:39 +0530
Initiating Ping Scan at 13:39
Scanning 185.230.63.171 [4 ports]
Completed Ping Scan at 13:39, 0.35s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 13:39
Completed Parallel DNS resolution of 1 host. at 13:39, 0.00s elapsed
DNS resolution of 1 IPs took 0.03s. Mode: Async [#: 2, OK: 1, NX: 0, DR: 0, SF: 0, TR: 1, CN: 0]
Initiating SYN Stealth Scan at 13:39
Scanning 171.63.230.185.in-addr.arpa (185.230.63.171) [1000 ports]
Discovered open port 80/tcp on 185.230.63.171
Discovered open port 445/tcp on 185.230.63.171
Discovered open port 443/tcp on 185.230.63.171
Discovered open port 82/tcp on 185.230.63.171
Completed SYN Stealth Scan at 13:39, 17.72s elapsed (1000 total ports)
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up, received syn-ack ttl 238 (0.27s latency).
Scanned at 2026-02-01 13:39:40 India Standard Time for 18s
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE      REASON
43/tcp    closed whois      reset ttl 242
80/tcp    open  http         syn-ack ttl 242
82/tcp    open  xfer         syn-ack ttl 242
83/tcp    closed mit-ml-dev  reset ttl 242
443/tcp   open  https        syn-ack ttl 242
445/tcp   open  microsoft-ds syn-ack ttl 238

Read data files from: C:\Program Files (x86)\Nmap
Nmap done: 1 IP address (1 host up) scanned in 18.56 seconds
Raw packets sent: 2007 (88.284KB) | Rcvd: 44 (1.924KB)

```

4. Increase debugging level (use -dd or more for greater effect)

```

C:\Users\Admin>nmap 185.230.63.171 -d
Packet.dll present, library version 1.83
wpcap.dll present, library version: Npcap version 1.83, based on libpcap version 1.10.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:38 +0530
PORTS: Using ports open on 0% or more average hosts (TCP:1000, UDP:0, SCTP:0)
----- Timing report -----
  hostgroups: min 1, max 100000
  rtt-timeouts: init 1000, min 100, max 10000
  max-scan-delay: TCP 1000, UDP 1000, SCTP 1000
  parallelism: min 0, max 0
  max-retries: 10, host-timeout: 0
  min-rate: 0, max-rate: 0
-----
Initiating Ping Scan at 13:38
Scanning 185.230.63.171 [4 ports]
Packet capture filter (device eth6): dst host 192.168.1.10 and (icmp or icmp6 or ((tcp or udp or sctp) and (src host 185.230.63.171)))
We got a TCP ping packet back from 185.230.63.171 port 443 (trynum = 0)
Completed Ping Scan at 13:38, 0.30s elapsed (1 total hosts)
Overall sending rates: 13.11 packets / s, 498.36 bytes / s.
mass_dns: Using DNS server fe80::1
mass_dns: Using DNS server 192.168.1.1
Initiating Parallel DNS resolution of 1 host. at 13:38
mass_dns: 0.03s 0/1 [#: 2, OK: 0, NX: 0, DR: 0, SF: 0, TR: 1]
Completed Parallel DNS resolution of 1 host. at 13:38, 0.01s elapsed
DNS resolution of 1 IPs took 0.03s. Mode: Async [#: 2, OK: 1, NX: 0, DR: 0, SF: 0, TR: 1, CN: 0]
Initiating SYN Stealth Scan at 13:38
Scanning 171.63.230.185.in-addr.arpa (185.230.63.171) [1000 ports]
Packet capture filter (device eth6): dst host 192.168.1.10 and (icmp or icmp6 or ((tcp or udp or sctp) and (src host 185.230.63.171)))
Discovered open port 443/tcp on 185.230.63.171
Discovered open port 80/tcp on 185.230.63.171
Discovered open port 445/tcp on 185.230.63.171
Discovered open port 82/tcp on 185.230.63.171
Completed SYN Stealth Scan at 13:38, 16.80s elapsed (1000 total ports)
Overall sending rates: 119.26 packets / s, 5247.51 bytes / s.
Nmap scan report for 171.63.230.185.in-addr.arpa (185.230.63.171)
Host is up, received syn-ack ttl 238 (0.26s latency).
Scanned at 2026-02-01 13:38:36 India Standard Time for 17s
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE REASON
43/tcp    closed whois  reset ttl 242
80/tcp    open  http  syn-ack ttl 237
82/tcp    open  xfer  syn-ack ttl 242
83/tcp    closed mit-ml-dev reset ttl 238
443/tcp    open  https  syn-ack ttl 242
445/tcp    open  microsoft-ds syn-ack ttl 242
Final times for host: srth: 262094 rttvar: 34262 to: 399142

Read from C:\Program Files (x86)\Nmap: nmap-protocols nmap-services.
Nmap done: 1 IP address (1 host up) scanned in 18.34 seconds
Raw packets sent: 2008 (88.312KB) | Rcvd: 35 (1.500KB)

```

5. Shows the host interfaces and routes

```

C:\Users\Admin>nmap 185.230.63.171 -iflist
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-01 13:37 +0530
*****INTERFACES*****
DEV (SHORT) IP/MASK TYPE UP MTU MAC
eth0 (eth0) fe80::d2af:c040:3ff4:15cc/64 ethernet down 1500 58:11:22:80:4E:82
eth0 (eth0) 169.254.21.246/16 ethernet down 1500 58:11:22:80:4E:82
unk0 (unk0) fd7a:115c:a1e0::f235:af76/128 other up 1280
unk0 (unk0) fe80::f569:f758:2df9:babf/64 other up 1280
unk0 (unk0) 100.86.175.118/32 other up 1280
eth1 (eth1) fe80::647e:6345:7e7f:3fa6/64 ethernet up 1500 0A:00:27:00:00:15
eth1 (eth1) 192.168.56.1/24 ethernet up 1500 0A:00:27:00:00:15
eth2 (eth2) fe80::1af0:8d77:167:b7fa/64 ethernet down 1500 F4:26:79:38:C3:91
eth2 (eth2) 169.254.239.202/16 ethernet down 1500 F4:26:79:38:C3:91
eth3 (eth3) fe80::d71b:2fd2:460f:7657/64 ethernet down 1500 F6:26:79:38:C3:90
eth3 (eth3) 169.254.250.255/16 ethernet down 1500 F6:26:79:38:C3:90
eth4 (eth4) fe80::7e05:689:789a:d364/64 ethernet up 1500 00:50:56:C0:00:01
eth4 (eth4) 192.168.64.1/24 ethernet up 1500 00:50:56:C0:00:01
eth5 (eth5) fe80::2d20:c19b:5e3b:1885/64 ethernet up 1500 00:50:56:C0:00:08
eth5 (eth5) 192.168.178.1/24 ethernet up 1500 00:50:56:C0:00:08
eth6 (eth6) 2402:a00:401:2590:a5bf:ec70:71c:7cca/64 ethernet up 1300 F4:26:79:38:C3:90
eth6 (eth6) 2402:a00:401:2590:8d9d:a250:530d:7dd1/128 ethernet up 1300 F4:26:79:38:C3:90
eth6 (eth6) fe80::342a:a448:dcf6:d035/64 ethernet up 1300 F4:26:79:38:C3:90
eth6 (eth6) 192.168.1.10/24 ethernet up 1300 F4:26:79:38:C3:90
lo0 (lo0) ::1/128 loopback up -1
lo0 (lo0) 127.0.0.1/8 loopback up -1

DEV WINDEVICE
eth0 \Device\NPF_{E9E3E001-C7F1-4010-9827-60432EFE2BCF}
eth0 \Device\NPF_{E9E3E001-C7F1-4010-9827-60432EFE2BCF}
unk0 \Device\NPF_{37217669-42DA-4657-A55B-0D995D328250}
unk0 \Device\NPF_{37217669-42DA-4657-A55B-0D995D328250}
unk0 \Device\NPF_{37217669-42DA-4657-A55B-0D995D328250}
eth1 \Device\NPF_{F45C22C5-FBBE-4546-8C1E-EF471363F577}
eth1 \Device\NPF_{F45C22C5-FBBE-4546-8C1E-EF471363F577}
eth2 \Device\NPF_{AC93695A-5973-4C5F-8DFB-3013A67A9B17}

```


Vulnerability Checking

```
(aastha@kali)-[~]
$ nmap --script vuln aasthathakker.com
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-27 05:14 EST
Nmap scan report for aasthathakker.com (185.230.63.171)
Host is up (0.21s latency).
Other addresses for aasthathakker.com (not scanned): 185.230.63.107 185.230.63.186
rDNS record for 185.230.63.171: unallocated.63.wixsite.com
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
|_ http-vuln-cve2014-3704: ERROR: Script execution failed (use -d to debug)
|_ http-vuln-wnr1000-creds: ERROR: Script execution failed (use -d to debug)
|_ http-aspnet-debug:
|_   status: DEBUG is enabled
|_ http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_ http-dombased-xss: Couldn't find any DOM based XSS.
|_ http-csrf: Couldn't find any CSRF vulnerabilities.
82/tcp    open  xfer
443/tcp   open  https
|_ http-csrf: Couldn't find any CSRF vulnerabilities.
|_ http-dombased-xss: Couldn't find any DOM based XSS.
|_ http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_ http-aspnet-debug:
|_   status: DEBUG is enabled
445/tcp   filtered microsoft-ds
2000/tcp  open  cisco-sccp
5060/tcp  open  sip

Nmap done: 1 IP address (1 host up) scanned in 346.57 seconds
```

```
(aastha@kali)-[~]
$ nmap --script ssl-enum-ciphers -p 443 aasthathakker.com
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-27 05:06 EST
Nmap scan report for aasthathakker.com (185.230.63.171)
Host is up (0.20s latency).
Other addresses for aasthathakker.com (not scanned): 185.230.63.107 185.230.63.186
rDNS record for 185.230.63.171: unallocated.63.wixsite.com

PORT      STATE SERVICE
443/tcp   open  https
|_ ssl-enum-ciphers:
|_   TLSv1.2:
|_     ciphers:
|_       TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (secp256r1) - A
|_       TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256 (secp256r1) - A
|_       TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 (secp256r1) - A
|_       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (secp256r1) - A
|_       TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA (secp256r1) - A
|_     compressors:
|_     cipher preference: server
|_   TLSv1.3:
|_     ciphers:
|_       TLS_AKE_WITH_AES_128_GCM_SHA256 (ecdh_x25519) - A
|_       TLS_AKE_WITH_AES_256_GCM_SHA384 (ecdh_x25519) - A
|_       TLS_AKE_WITH_CHACHA20_POLY1305_SHA256 (ecdh_x25519) - A
|_     cipher preference: server
|_   least strength: A

Nmap done: 1 IP address (1 host up) scanned in 20.72 seconds
```

80,443 - nmap without privileges

icmp-echo, tcp-syn 80, icmp time stamp, tcp ack 443 - nmap with root

